

Sri Lanka Institute of Information Technology



Data Warehousing and Business Intelligence - IT3021 Assignment

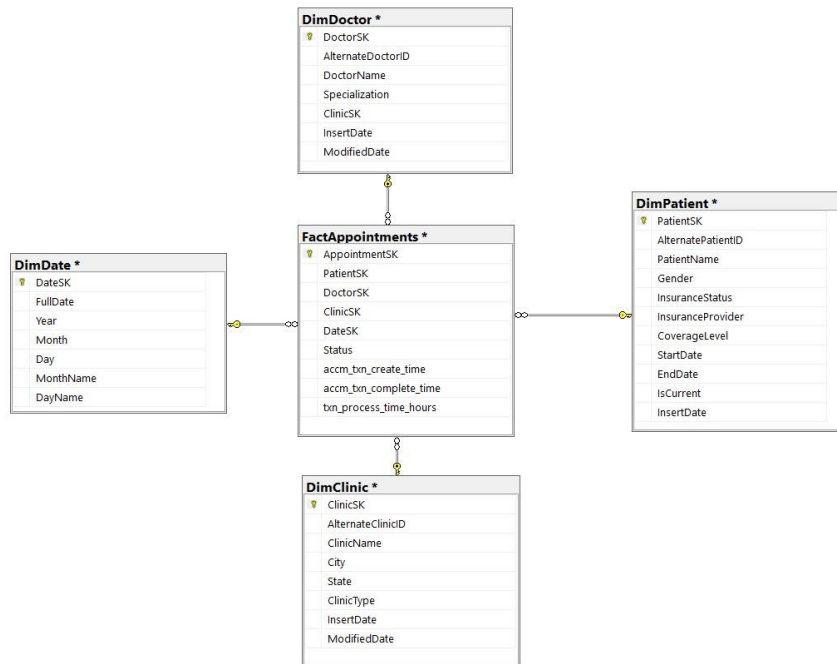
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Step 1: Data source for the Assignment 2



This Medical Center Data Warehouse (DW) is designed using a star schema to support appointment analysis and healthcare decision-making. The central fact table, FactAppointments, stores key transactional data related to appointments, including patient, doctor, clinic, and date references, along with processing time metrics.

The fact table is linked to multiple dimension tables that provide descriptive details. DimPatient contains patient information such as gender and insurance details and is implemented as a Slowly Changing Dimension Type 2 (SCD Type 2) to maintain historical records. DimDoctor includes doctor details such as name and specialization, while DimClinic provides information about clinics, including location and clinic type.

DimDate enables time-based analysis with attributes such as year and month, supporting trend and performance analysis over time.

All dimensions are connected to the fact table through surrogate keys, forming a star schema that supports efficient querying and analytical reporting using OLAP and Power BI tools.

Step 2: SSAS Cube implementation

1. The SSAS cube was developed using SQL Server Data Tools (SSDT) to enable multidimensional analysis on the MedicalCenter data warehouse.
2. A data source was then configured by connecting to the MedicalCenter_DW database using Windows authentication

The screenshot displays the 'Data Source Designer' window, which is used for configuring data sources in SQL Server Analysis Services (SSAS). The window is titled 'Data Source Designer' and has two tabs: 'General' and 'Impersonation Information'. The 'General' tab is currently selected.

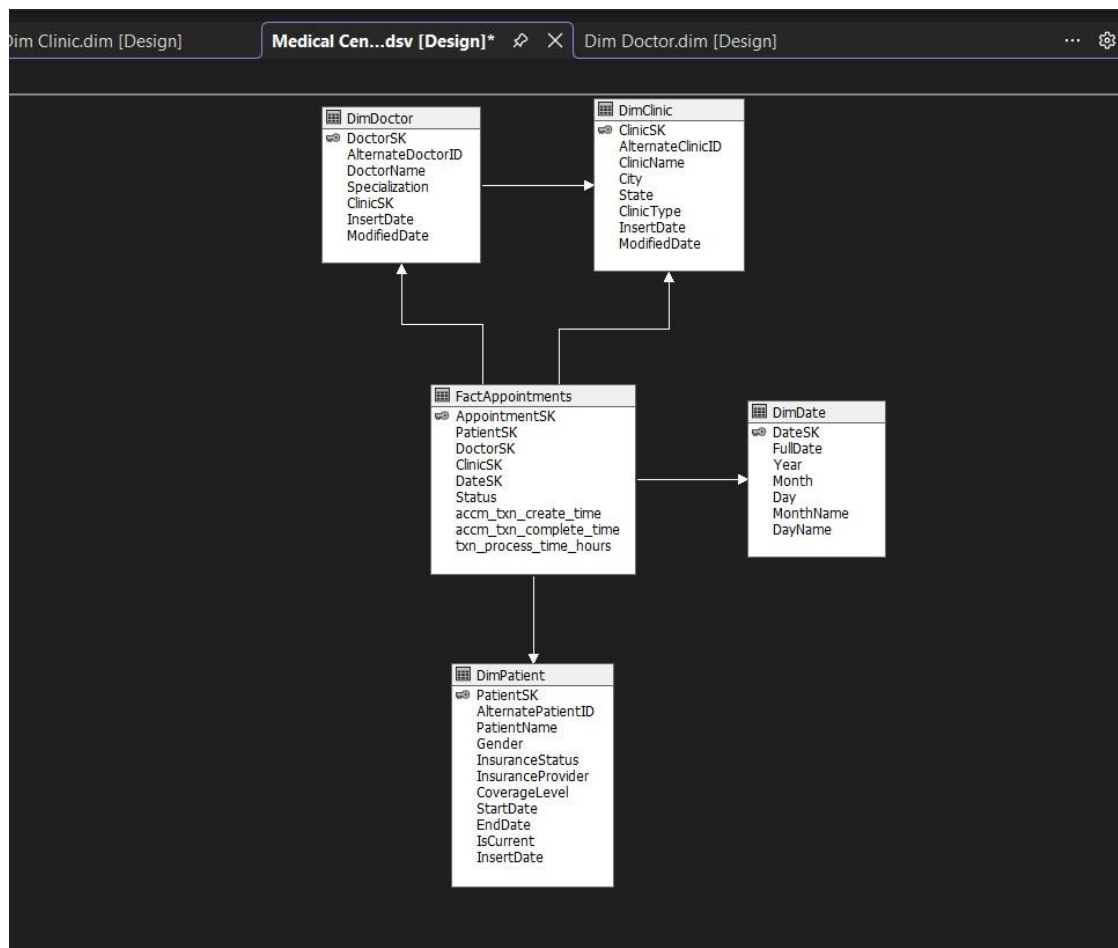
Key configuration details visible in the 'General' tab include:

- Data source name:** Medical Center DW
- Provider:** SQLOLEDB.1
- Connection string:** Provider=SQLOLEDB.1;Data Source=DESKTOP-MMSDV4Q
- Isolation:** ReadCommitted
- Query timeout (in seconds):** 0
- Maximum number of connections:** 10
- Data source description:** (Empty text area)

The 'Data source references' section contains a checkbox labeled 'Maintain a reference to another object in the solution', which is currently unchecked. Below this checkbox is a dropdown menu set to 'Create a data source based on an existing data source'.

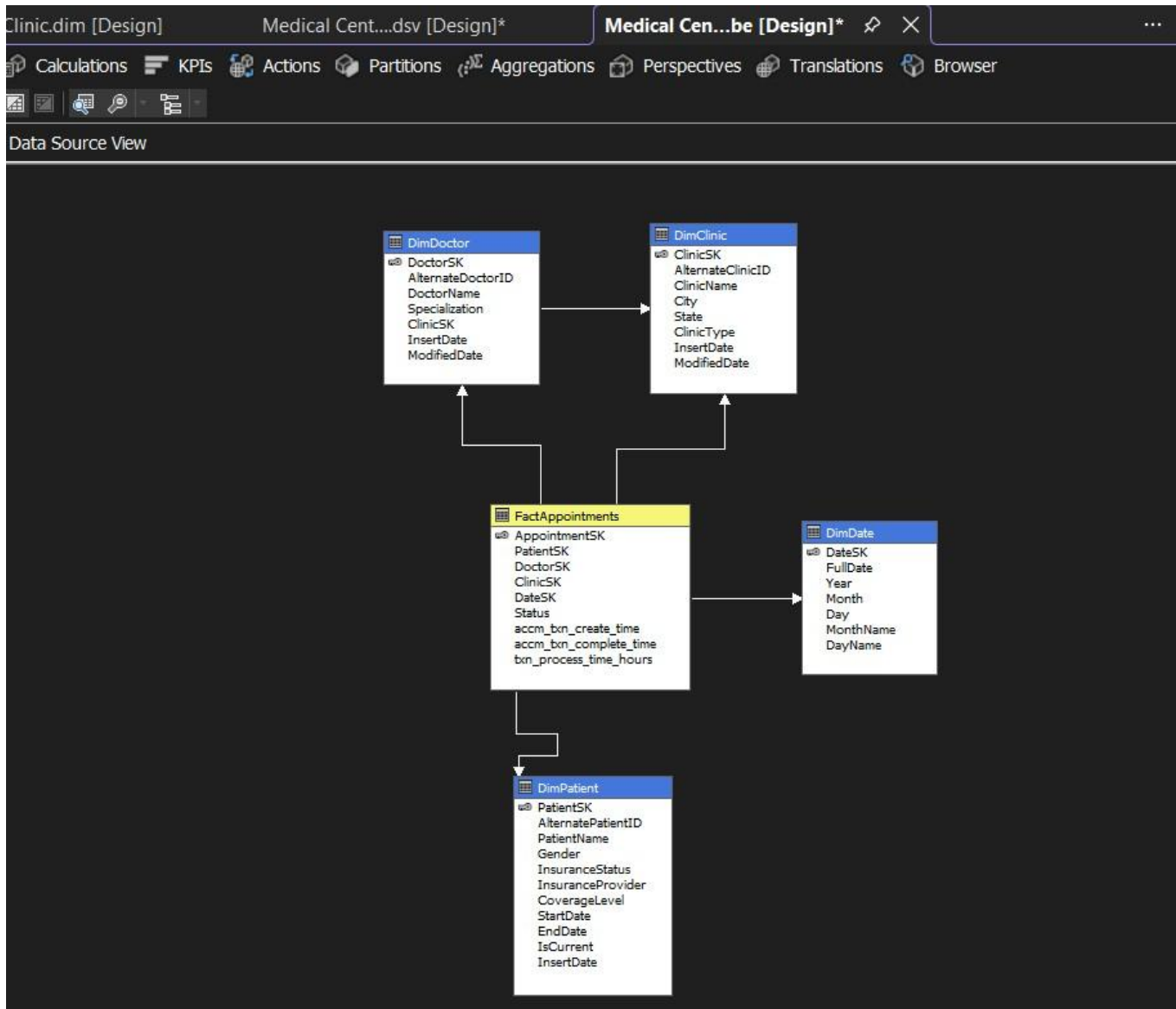
At the bottom of the window, there are three buttons: 'OK', 'Cancel', and 'Help'.

3. A Data Source View (DSV) was created by importing the required tables, including the fact table (FactAppointments) and the related dimension tables (DimPatient, DimDoctor, DimClinic, and DimDate). Relationships between these tables were verified within the DSV to ensure correct data mapping.

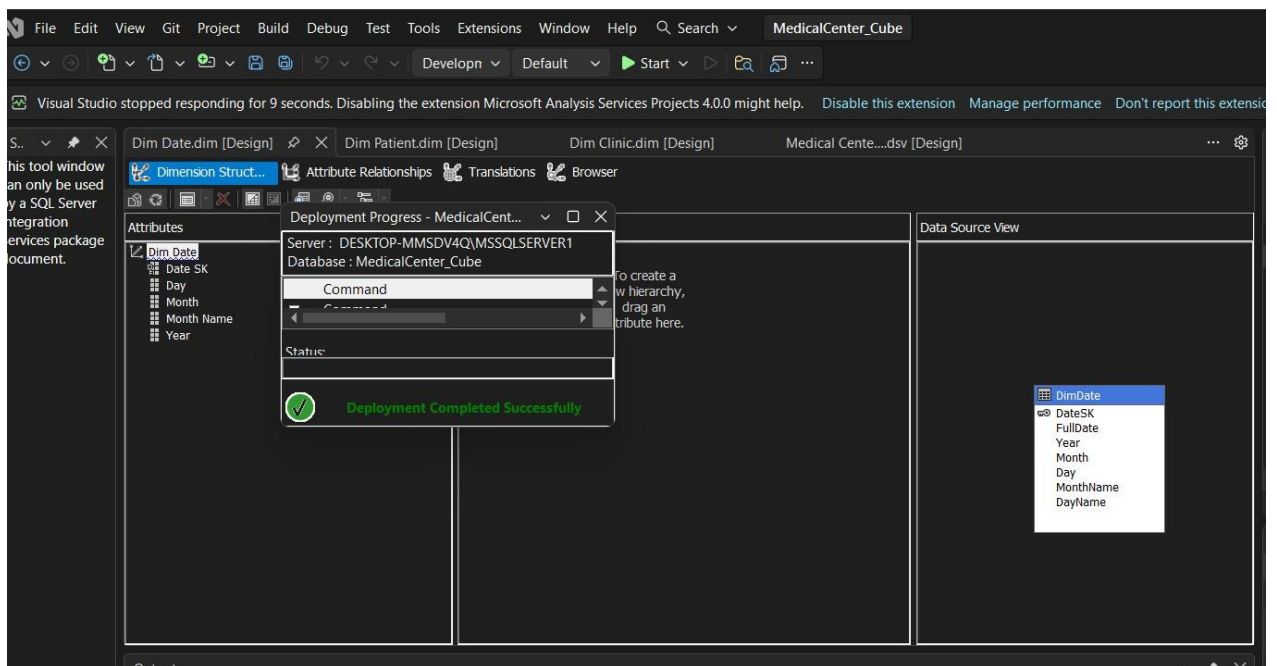
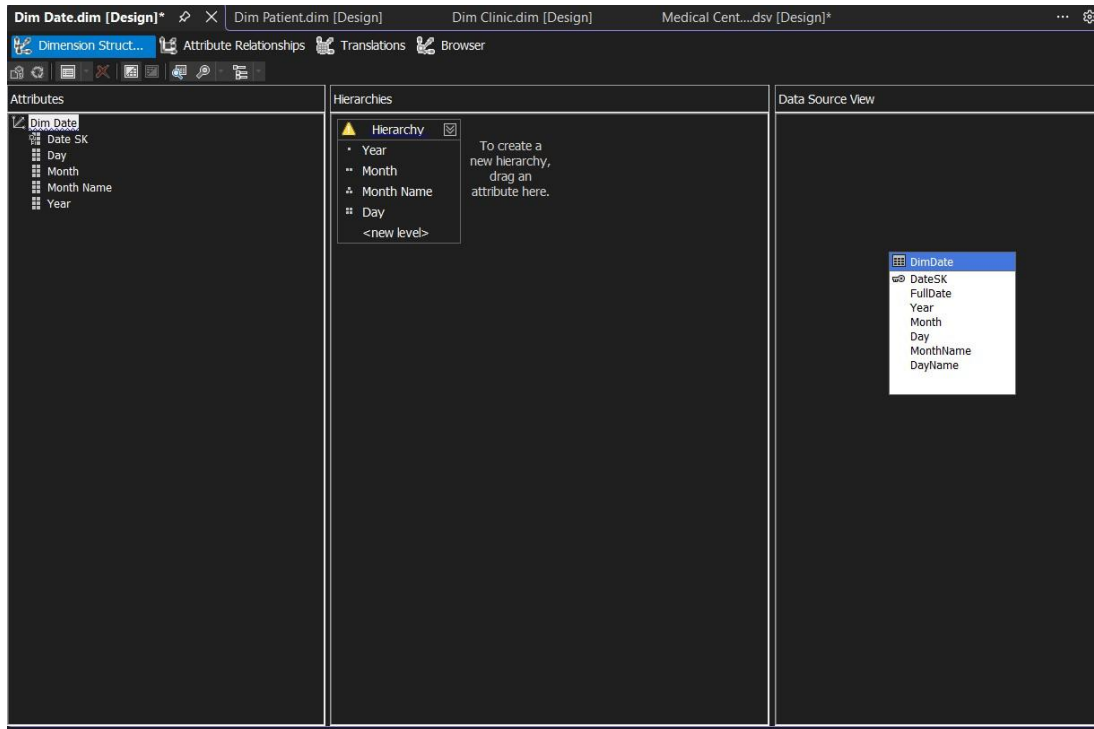


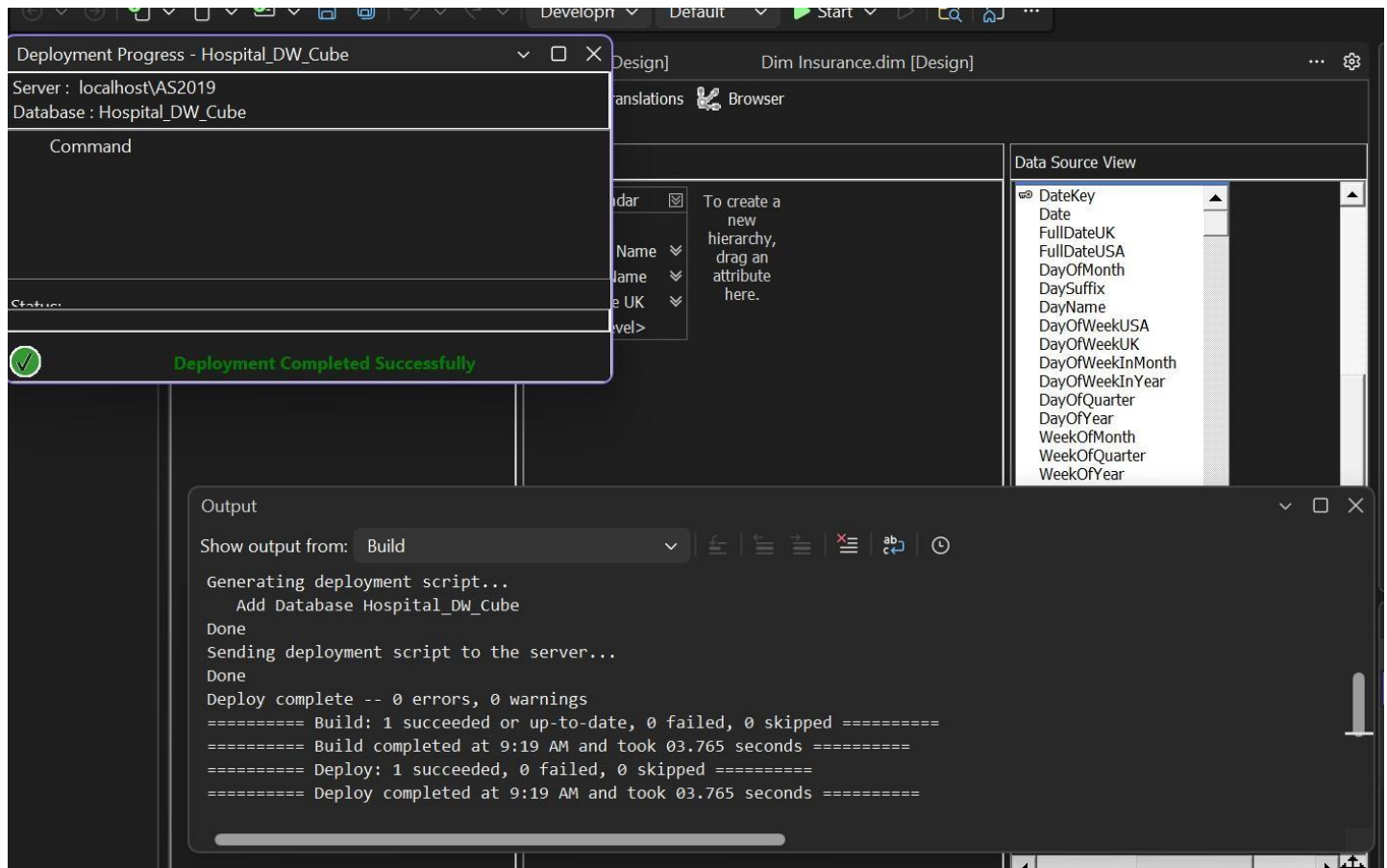
4. The Cube Wizard was used to create the cube by selecting the "Use existing tables" option. The FactAppointments table was selected as the measure group, and all dimension tables were added.
5. Each dimension was then configured by selecting appropriate attributes and defining key fields. Additionally, a hierarchy was created in the DimDate dimension (Year → Month → Day) to support drill-down and time-based analysis.

- The cube was deployed to the SSAS server and processed successfully. The deployment was validated by browsing the cube in SSDT and verifying accessibility through SQL Server Management Studio.



Date Hierarchy → Year – Month – Month name - Day





Step 3: Demonstration of OLAP operations

1. Connected Excel to SSAS Cube – Microsoft Excel was connected to the deployed SSAS cube using the *Data → Get Data → From Analysis Services* option.
2. Created PivotTable – A PivotTable was created using the cube data to perform multidimensional analysis by dragging dimensions and measures into rows, columns, and values

Data Connection Wizard

Select Database and Table

Select the Database and Table/Cube which contains the data you want.

Select the database that contains the data you want:

MedicalCenter_Cube

☒ Connect to a specific cube or table:

Name	Description	Modified	Created	Type
 Medical Center DW		4/3/2026 8:43:22 PM		CUBE

Cancel < Back Next > Finish

- Demonstrated Roll-up – Data was aggregated from a detailed level to a higher level by summarizing total appointments by Year

Year	Appointments Count
2025	467466
2026	132534
Grand Total	600000

- Performed Drill-down – Hierarchical analysis was carried out by expanding data from **Year to Month level** to view more detailed information.

Year	Appointments Count
2025	
March	16710
April	49386
May	50292
June	48216
July	50712
August	51444
September	48672
October	50226
November	49986
December	51822
2026	
January	50958
February	45072
March	36504
Grand Total	600000

5. Applied Slice Operation – Data was filtered using a single dimension, such as selecting a specific Clinic Type (Private/Public)

Clinic Type	Private
Year	Appointments Count
2025	
March	6792
April	19764
May	20124
June	19428
July	20694
August	20916
September	19698
October	19938
November	19728
December	20814
2026	
January	20316
February	17574
March	14820
Grand Total	240606

Clinic Type	Public
Year	Appointments Count
2025	
March	9918
April	29622
May	30168
June	28788
July	30018
August	30528
September	28974
October	30288
November	30258
December	31008
2026	
January	30642
February	27498
March	21684
Grand Total	359394

6. Executed Dice Operation – Multiple filters were applied simultaneously, such as Year and Clinic Type, to analyze a subset of the data.

Year	2025
Clinic Type	Public
Year	Appointments Count
March	9918
April	29622
May	30168
June	28788
July	30018
August	30528
September	28974
October	30288
November	30258
December	31008
Grand Total	279570

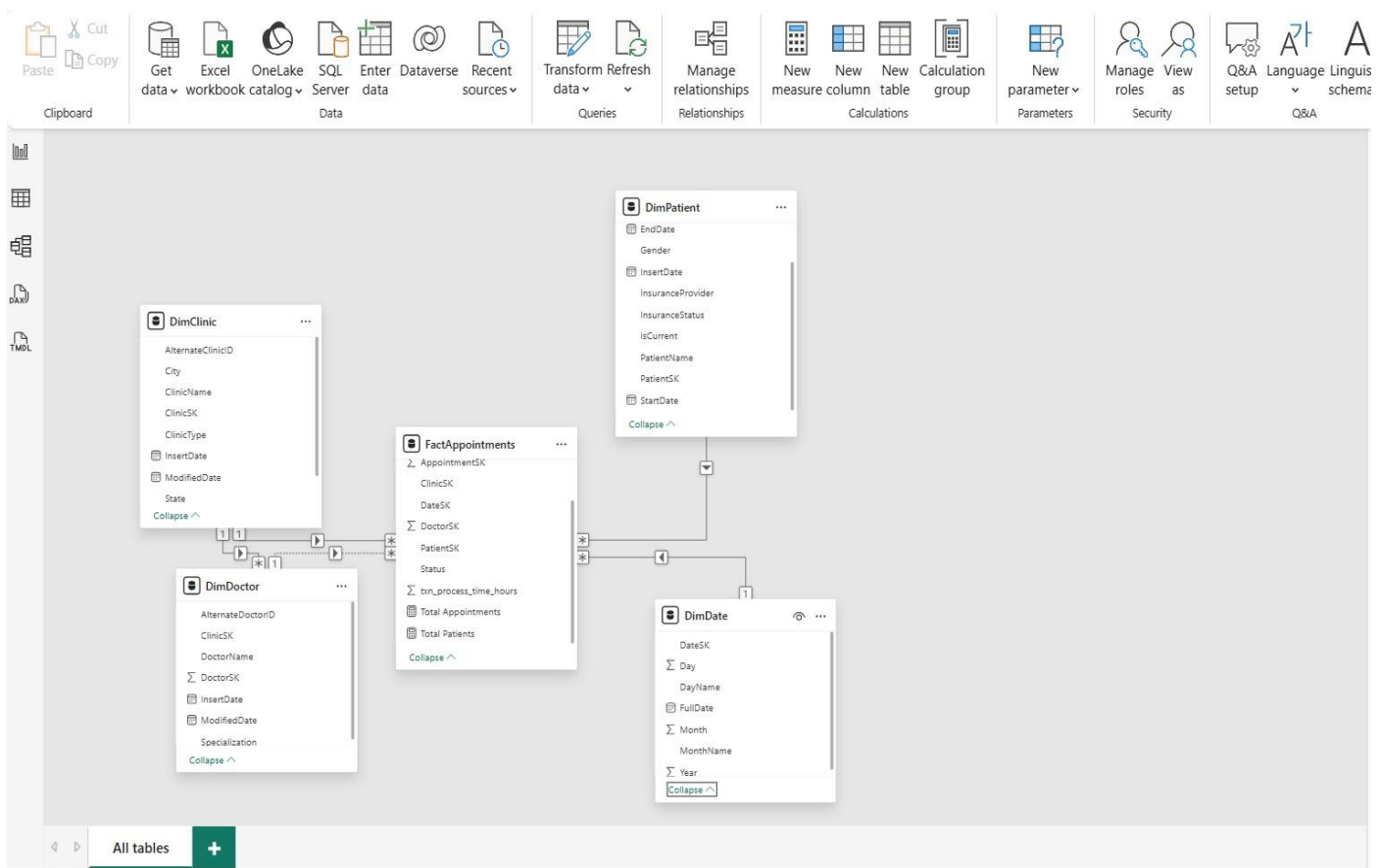
7. Showed Pivot Functionality – Rows and columns were rearranged (e.g., Clinic Type in columns and Month in rows) to view data from different perspectives.

Appointments Count	Clinic Type	
Year	Private	Public
2025	187896	279570
2026	52710	79824
Grand Total	240606	359394

Appointments Count	Clinic Type		
Year	Private	Public	Grand Total
2025			
March	6792	9918	16710
April	19764	29622	49386
May	20124	30168	50292
June	19428	28788	48216
July	20694	30018	50712
August	20916	30528	51444
September	19698	28974	48672
October	19938	30288	50226
November	19728	30258	49986
December	20814	31008	51822
2026			
January	20316	30642	50958
February	17574	27498	45072
March	14820	21684	36504
Grand Total	240606	359394	600000

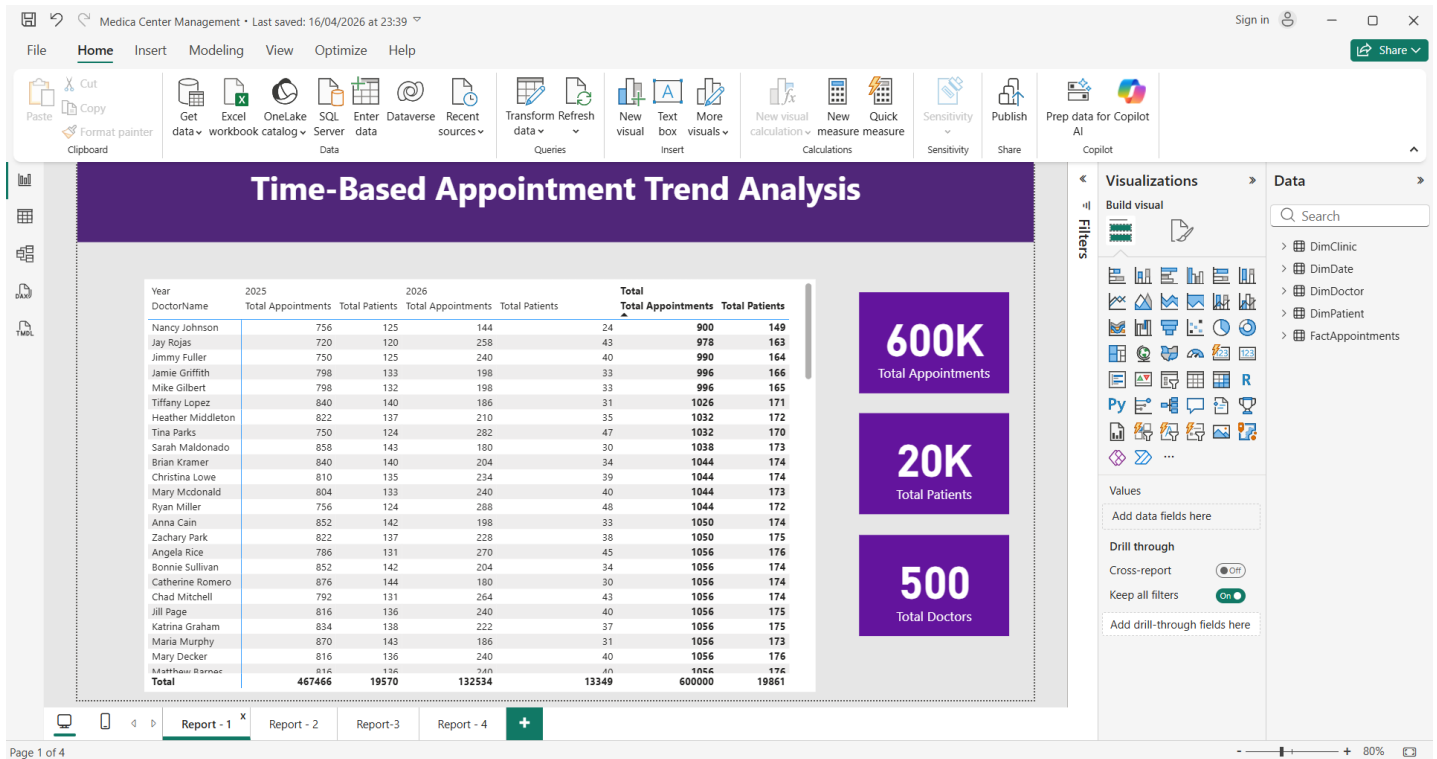
Step 4: PowerBI Reports

1. Power BI Desktop was used to create reports by connecting to the MedicalCenter_DW database. The required tables (FactAppointments, DimPatient, DimDoctor, DimClinic, and DimDate) were imported and relationships were established to form a star schema. DAX measures such as Total Appointments, Total Patients, and Average Processing Time were created to support analysis.



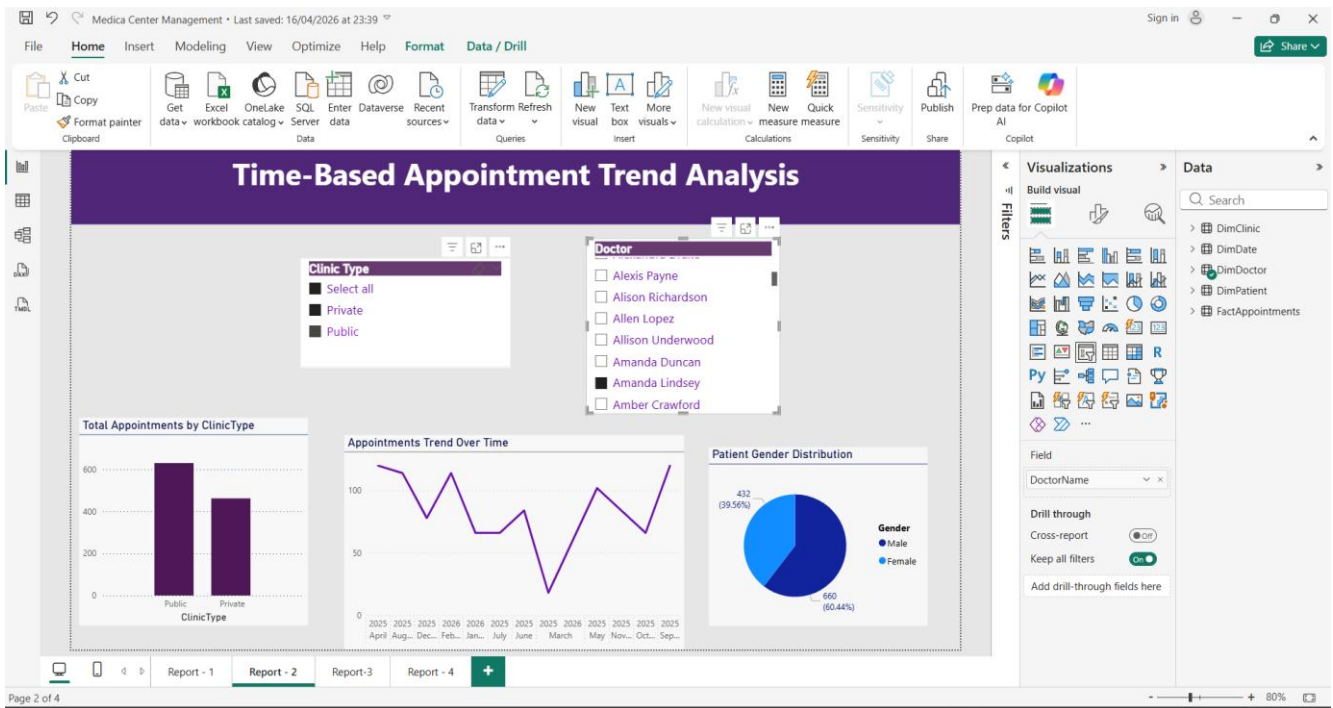
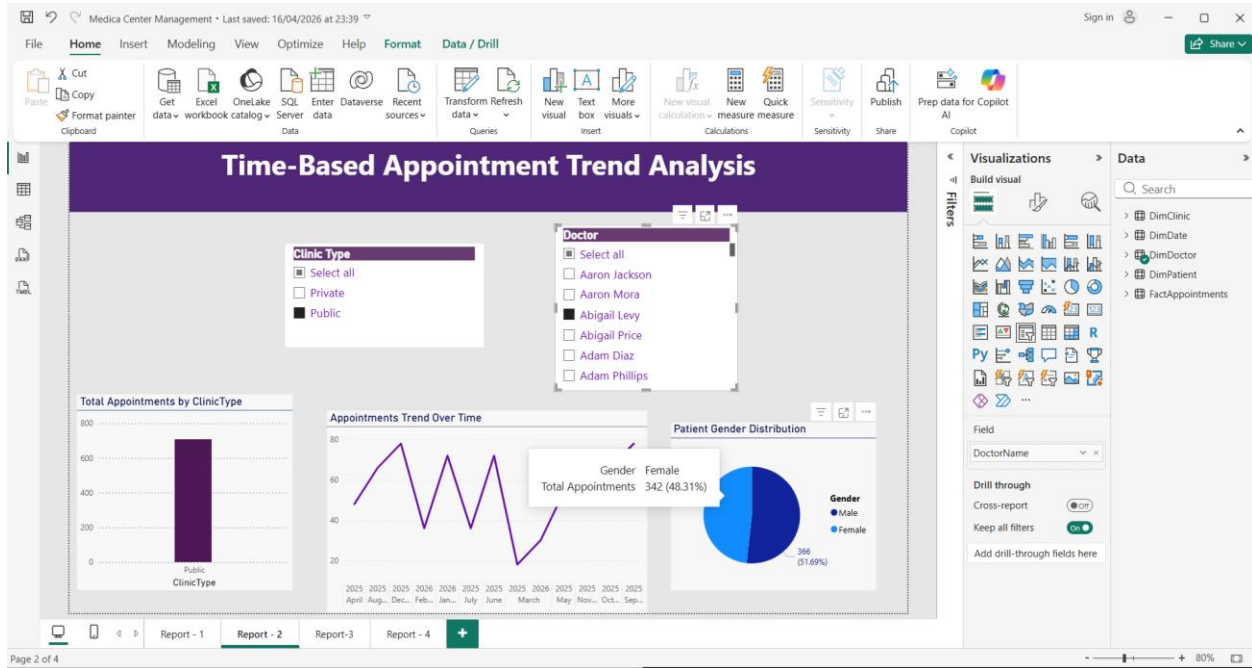
Report 1: Matrix Visual

A matrix visual was designed to display detailed tabular data. Doctor Name was used as rows and Year as columns, while Total Appointments was used as the value. This report provides a clear comparison of appointment counts across different doctors and time periods



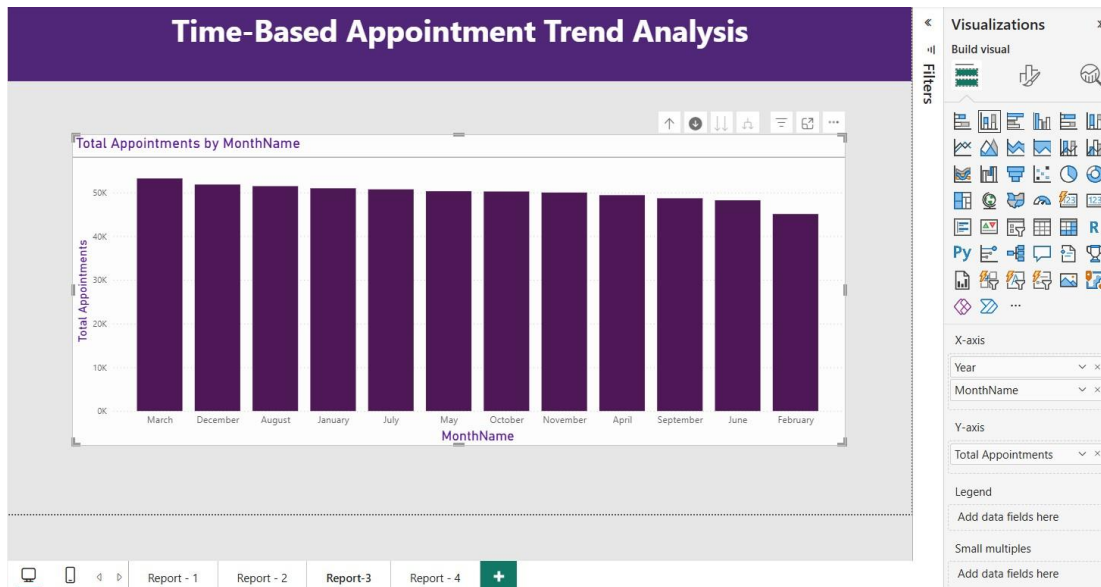
Report 2: Multiple Slicers with Cascading Filters

An interactive dashboard was created using multiple slicers. Clinic Type and Doctor Name were used as slicers, where selecting a clinic type dynamically filters the available doctors (cascading filter). Additional visuals such as bar charts, pie charts, and line charts were included to present insights from different perspectives.



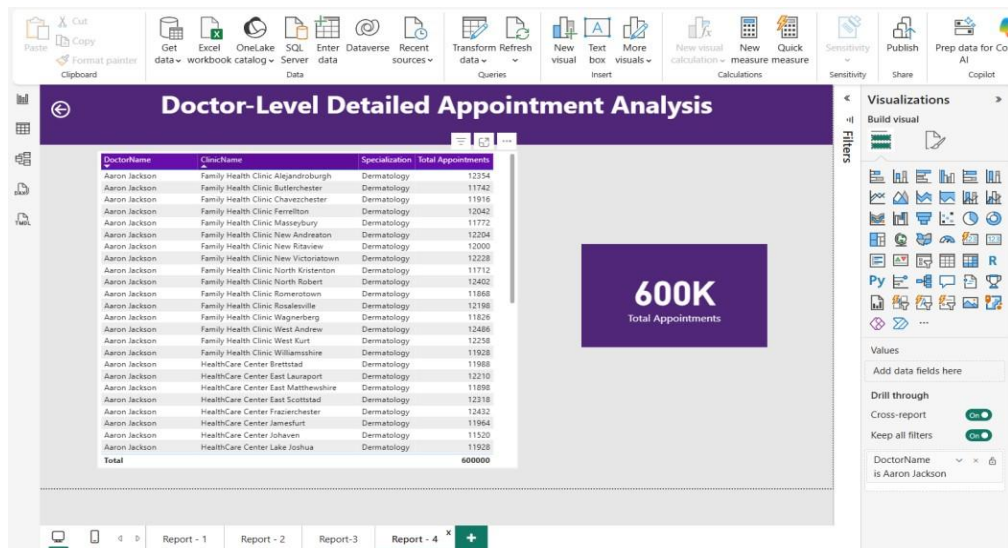
Report 3: Drill-Down Analysis

A column chart was created with a time hierarchy (**Year** → **Month**) to enable drill-down functionality. This allows users to move from a high-level overview to more detailed insights and analyze trends over time.



Report 4: Drill-Through Report

A separate drill-through page was created to provide detailed information. **Doctor Name** was used as the drill-through field, allowing users to right-click on a doctor in another report and navigate to a detailed view of appointment data related to that doctor.



DAX Measures

Total Appointments = COUNT(FactAppointments[AppointmentSK])

Total Patients = DISTINCTCOUNT(FactAppointments[PatientSK])

Total Doctors = DISTINCTCOUNT(DimDoctor[DoctorSK])

Power BI Dashboard

[Link for the Power BI Dashboard](#)